

CHAPTER 5 • FOR PHYSIOTHERAPY

When the vessel breaks

Pathophysiology of non-traumatic intracerebral haemorrhage

Chapter 5: Rossi and Cordonnier (2014), pp. 51–60

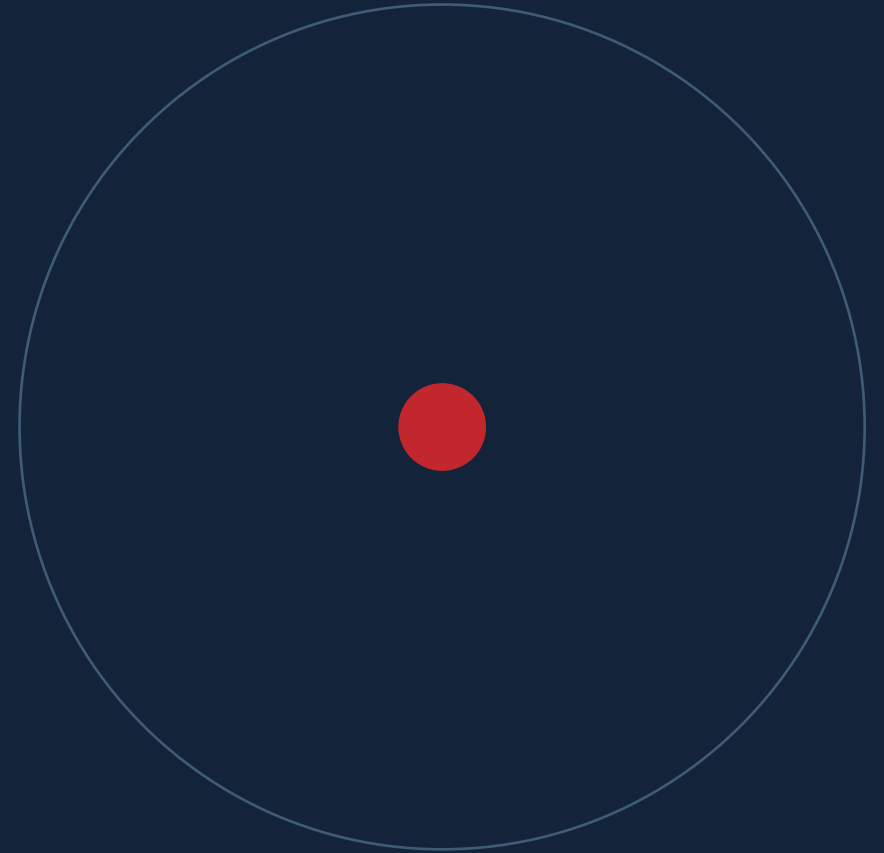
In B. Norrving (Ed.), *Oxford textbook of stroke and cerebrovascular disease*.
Oxford University Press.



10 December 1996

7 a.m. · Jill Bolte Taylor, neuroanatomist, aged 37

“I woke up to a pounding pain behind my left eye.”



Source: Taylor (2008a).

10 December 1996

7 a.m. · Jill Bolte Taylor, neuroanatomist, aged 37

Within four hours she could not **walk, talk, read or write.**



Source: Taylor (2008a).

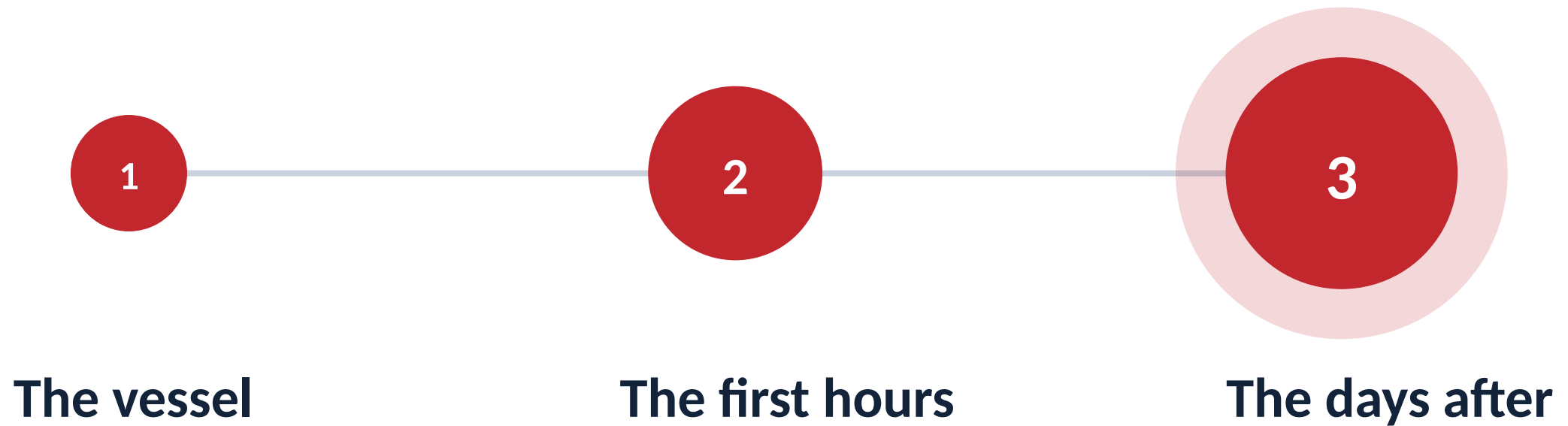
40%

**die within a month of an
intracerebral haemorrhage**

No improvement from 1980 to 2008.

Median 1-month case fatality 40.4%; 36 population studies, n = 8145 (van Asch et al., 2010).

ICH is a process, not a moment



Each stage is a window to act.

Framework adapted from Rossi and Cordonnier (2014).

Four things you will be able to do

Explain

why a bleed keeps injuring the brain

Predict

the impairment and the risk of decline

Appraise

the strength of each claim

Apply

time and pace therapy to the stage

Two ways to listen

THE TITLE

is the message. New to stroke?
Hold on to it.

THE TAG

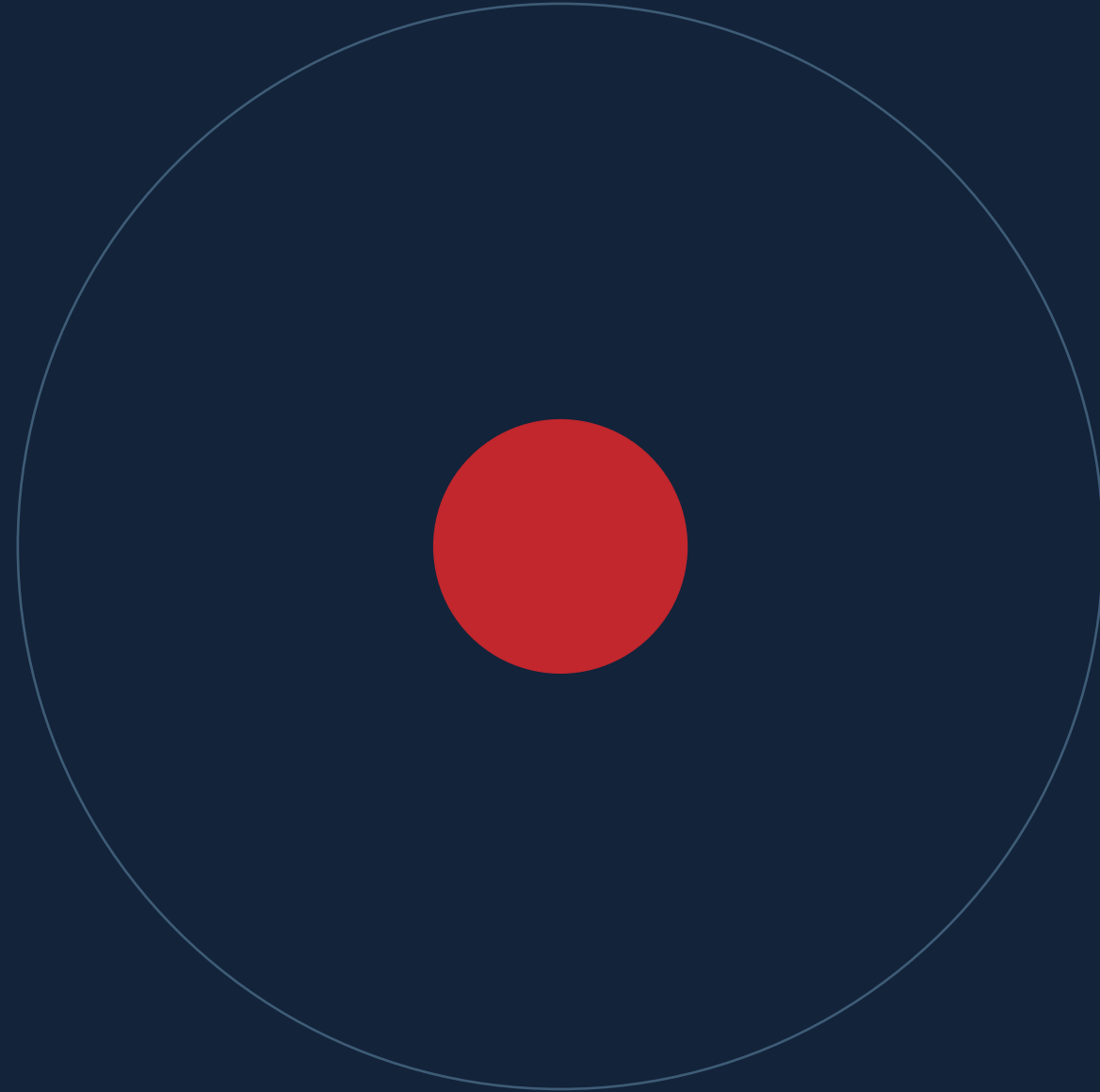
🔍 Cohort • n = 1013

is the evidence. Experienced?
Challenge it.

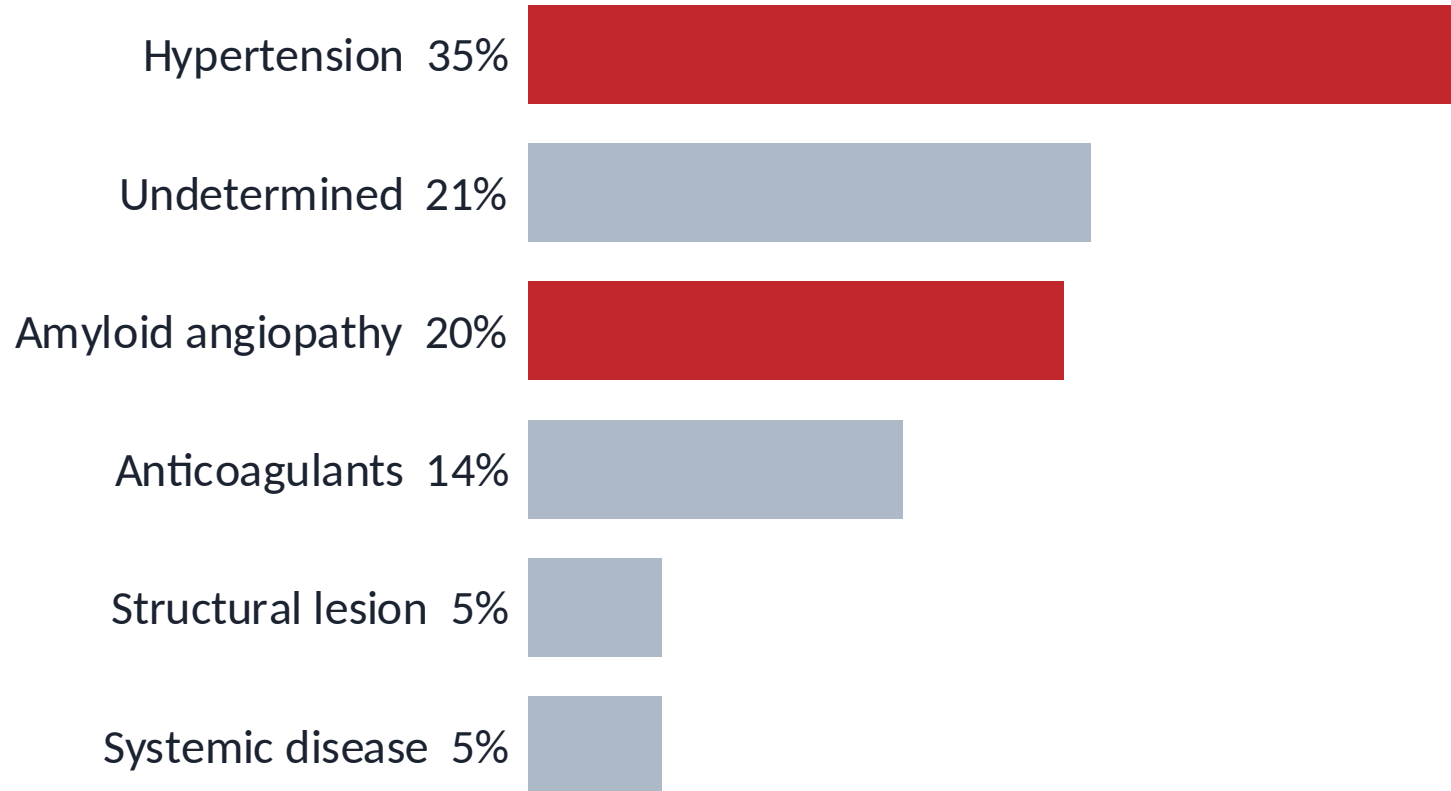
01

The vessel

Why does a small artery give way?



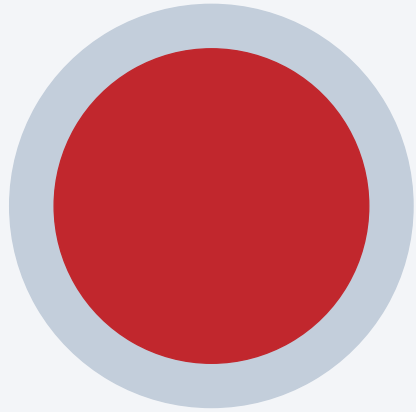
Hypertension and amyloid cause over half



55%
two small-vessel
diseases

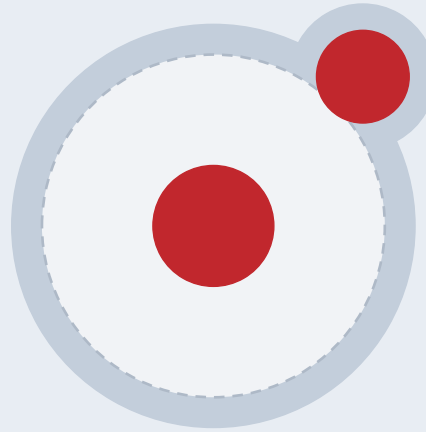
SMASH-U classification, n = 1013, Helsinki (Meretoja et al., 2012). Full breakdown on the handout.

Two diseases, two sets of vessels



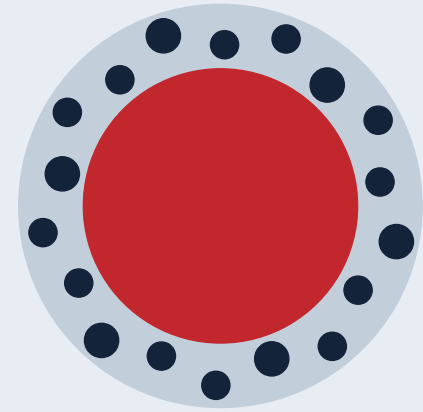
Healthy

Thin wall, wide channel



Hypertension

Deep arteries: thick, brittle wall

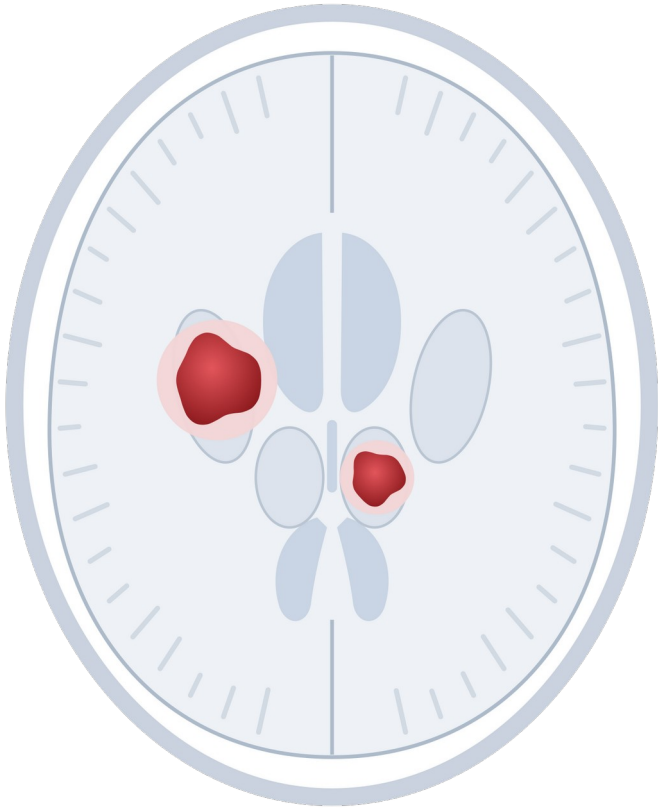


Amyloid angiopathy

Surface arteries: amyloid replaces muscle

Hypertensive arteriopathy (Fisher, 1971); amyloid angiopathy (Rossi & Cordonnier, 2014).

Where it bleeds tells you why



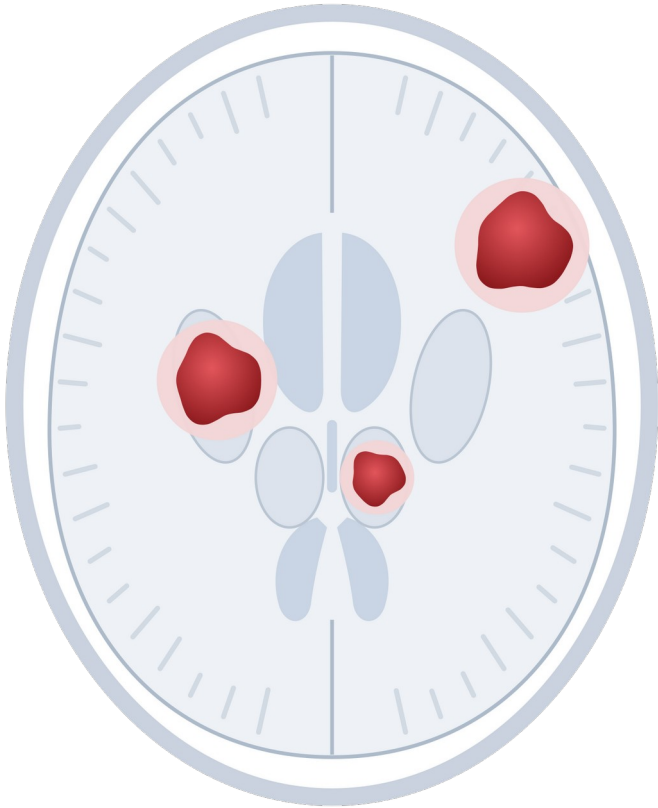
Deep

→ hypertension

putamen • thalamus • pons • cerebellum

Schematic axial slice; the pons and cerebellum sit below this level. Deep vs lobar patterns (Macellari et al., 2014; Meretoja et al., 2012).

Where it bleeds tells you why



Deep

→ hypertension

putamen • thalamus • pons • cerebellum

Lobar

→ amyloid (older adults)

Young or atypical? Look for a lesion.

Schematic axial slice; the pons and cerebellum sit below this level. Deep vs lobar patterns (Macellari et al., 2014; Meretoja et al., 2012).

Location predicts the impairment

Putamen, internal capsule Opposite-side weakness. **Clot on the PLIC: worse motor outcome.**

Thalamus Opposite-side sensory loss, ataxia; later, central pain

Cerebellum Same-side ataxia, truncal sway. Drowsy = emergency.

Pons Often severe: four-limb weakness, cranial nerve signs

Lobar By lobe: weakness, neglect, aphasia, field loss; seizures

PLIC involvement and motor outcome (Gupta et al., 2025; Murray et al., 2025; Puig et al., 2019).

Microbleeds are the footprints



Surface
→ amyloid



Deep
→ hypertension

60%

**of people with ICH have
microbleeds on MRI**

Healthy adults: 5%

Systematic review (Cordonnier et al., 2007). Boston criteria on the handout (Knudsen et al., 2001).

Jill: primary or secondary?



37 years old



No hypertension



Lobar bleed



Case details: Taylor (2008b).

Secondary: an arteriovenous malformation



37 years old



No hypertension



Lobar bleed

65%

of patients like Jill had a lesion on angiography

0%

if over 45, hypertensive, with a deep bleed

Prospective; 206 patients, all had catheter angiography (Zhu et al., 1997). Other secondary causes on the handout.

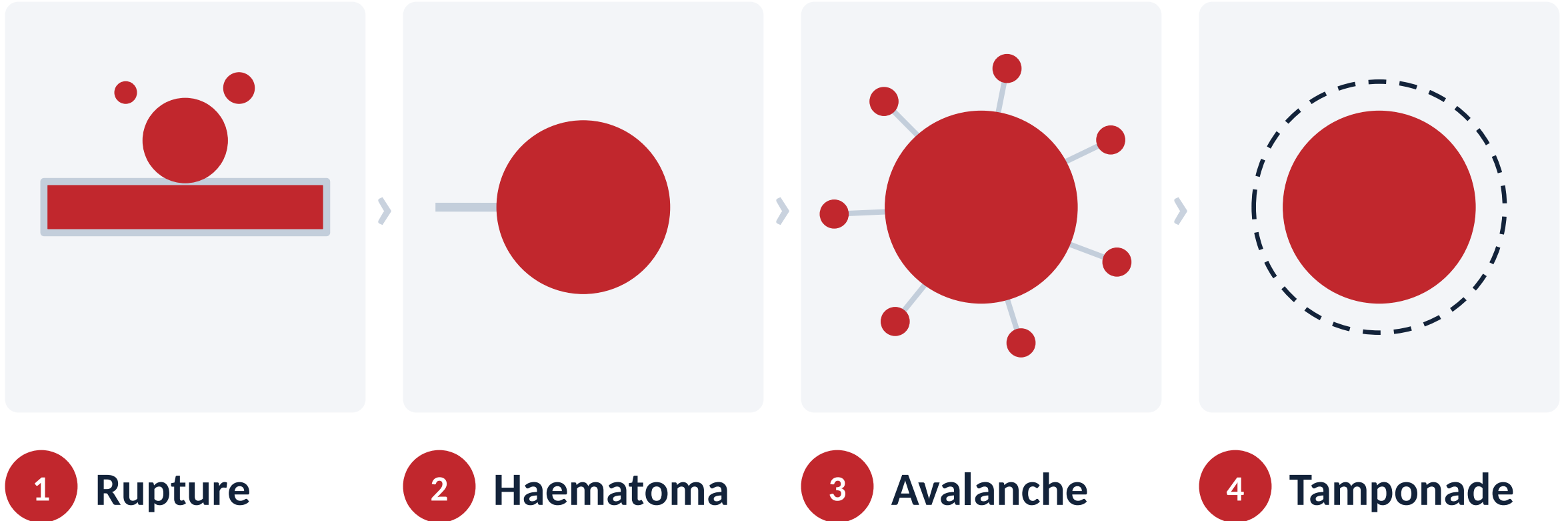
02

The first hours

Rupture, growth and pressure

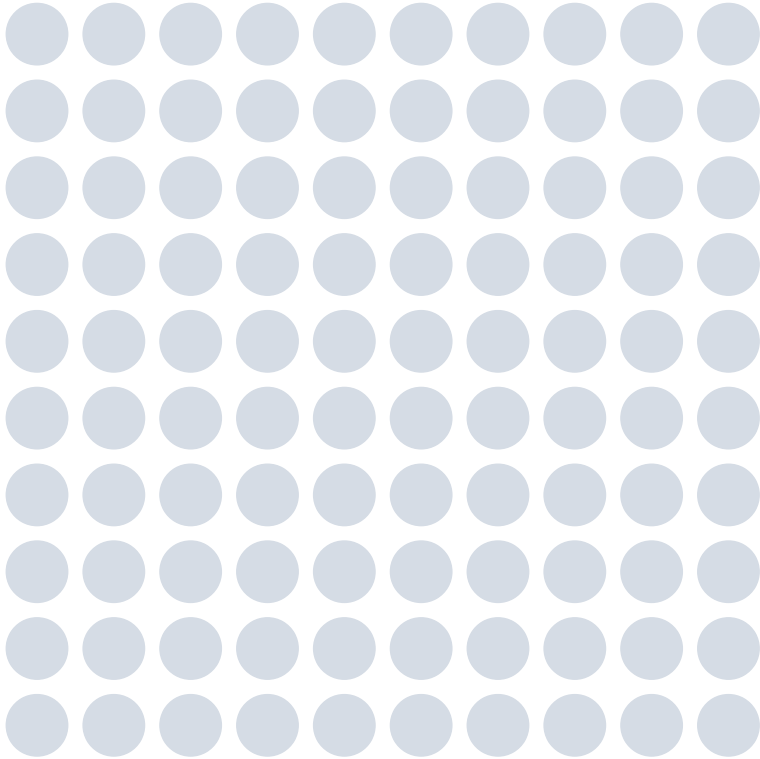


One rupture can trigger more



Secondary bleeding points at the edge of the haematoma (Fisher, 1971).

How many bleeds keep growing?



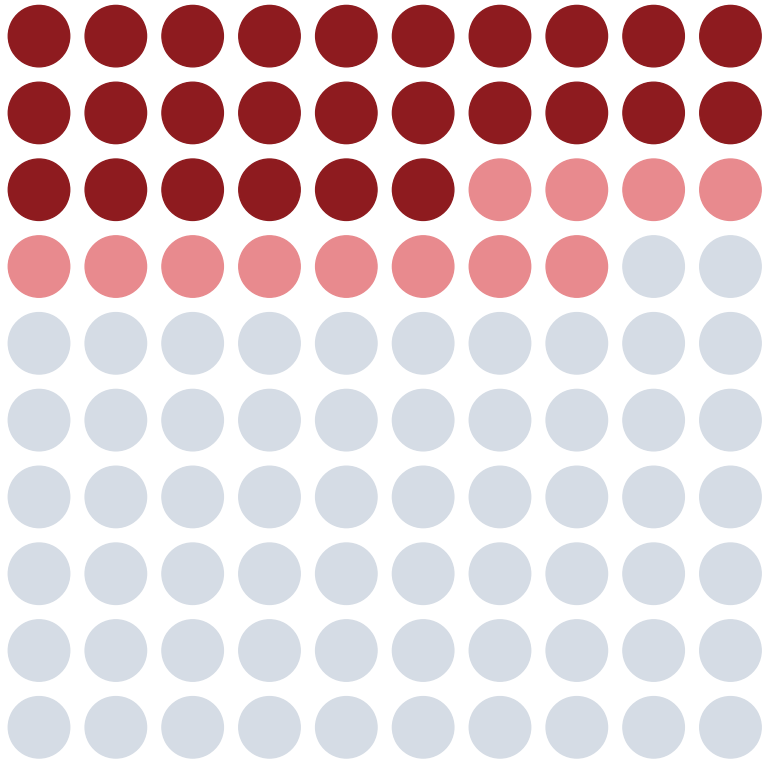
Each dot = one patient, scanned within 3 hours of onset

A 5 in 100

B 15 in 100

C 40 in 100

38 in 100 keep growing



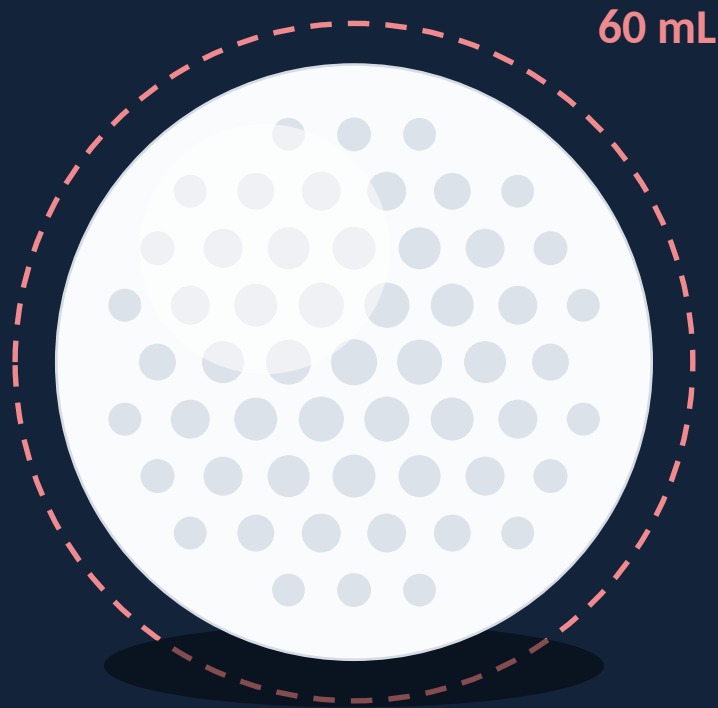
Each dot = one patient, scanned within 3 hours of onset

- 26 grew within the first hour
- 12 more grew within 20 hours
- 62 did not grow substantially

Jill: four hours
from headache to losing her speech

Growth > 33%, n = 103 (Brott et al., 1997). Every 10% of growth: +5% hazard of death (Davis et al., 2006).

Jill's clot was the size of a golf ball



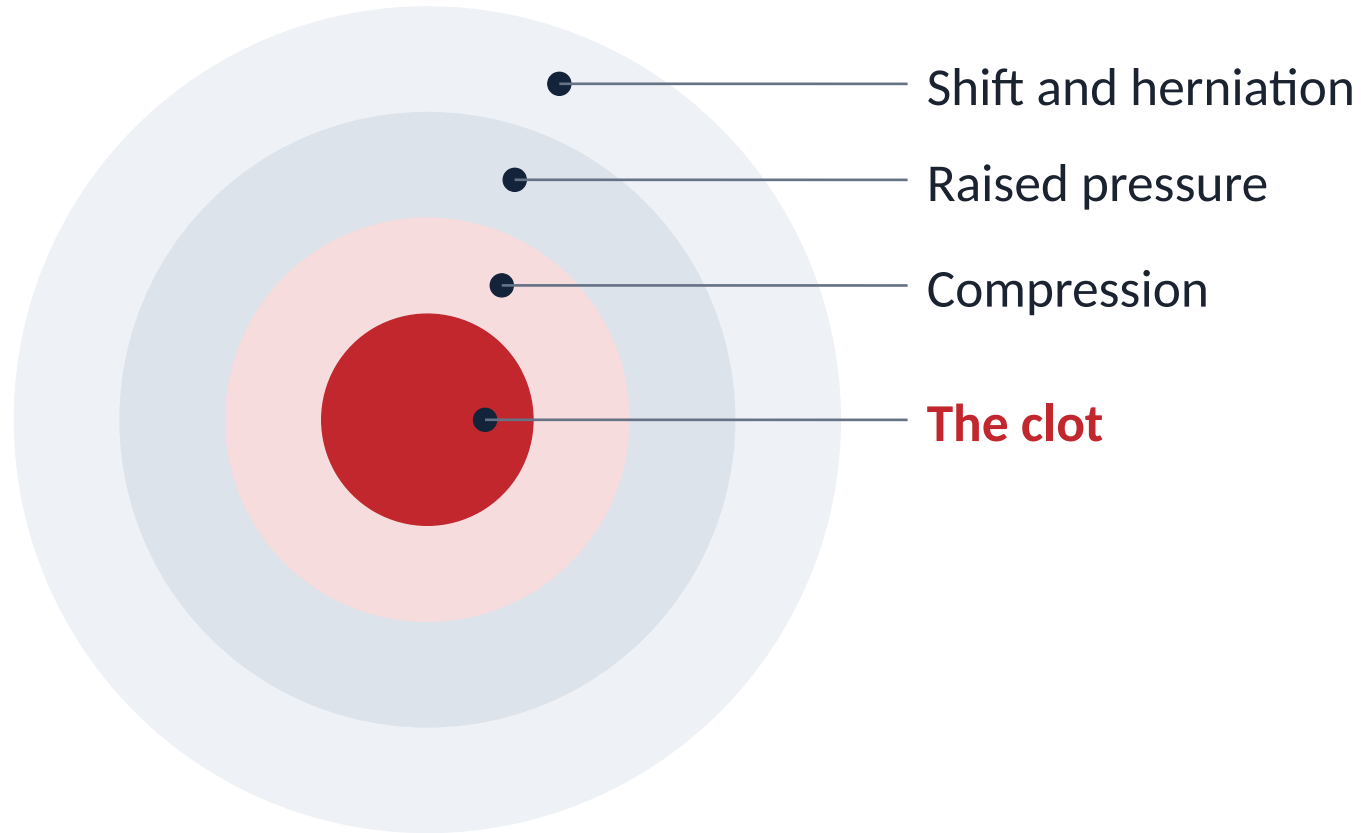
A golf ball $\approx$ 40 mL

60 mL + GCS $\leq$ 8: **91% predicted 30-day mortality**

A clot that grows by a third looks only 10% wider on a scan.

Volume and 30-day mortality, n = 188 (Broderick et al., 1993). Clot size (Taylor, 2008b). A golf ball is 42.7 mm across (about 40.7 mL).

The skull is a closed box



≈45%

break into the ventricles

→ hydrocephalus, worse outcome

Intraventricular extension and outcome (Hanley, 2009).

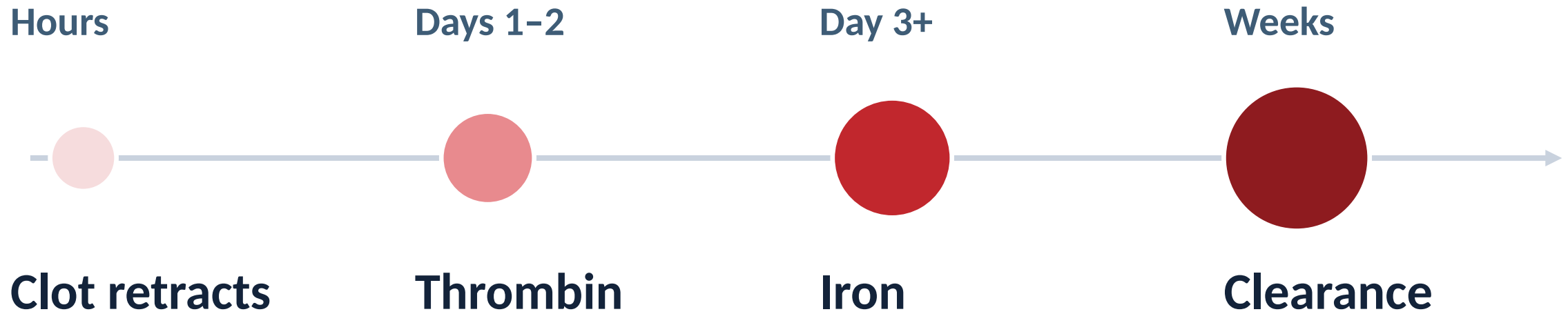
03

The days after

When the blood becomes a toxin



Oedema builds in waves



+75% oedema in the first 24 hours

Jill: surgery 17 days later

Oedema mechanisms (Xi et al., 2006); +75% in 24 h (Gebel et al., 2002); timing of Jill's surgery (Taylor, 2008b).

Three toxins, one result



Thrombin



Iron



Inflammation



Leaky barrier → more oedema → neuronal death

Mechanisms (Xi et al., 2006). The deferoxamine trial i-DEF was neutral (Selim et al., 2019).

True or false?

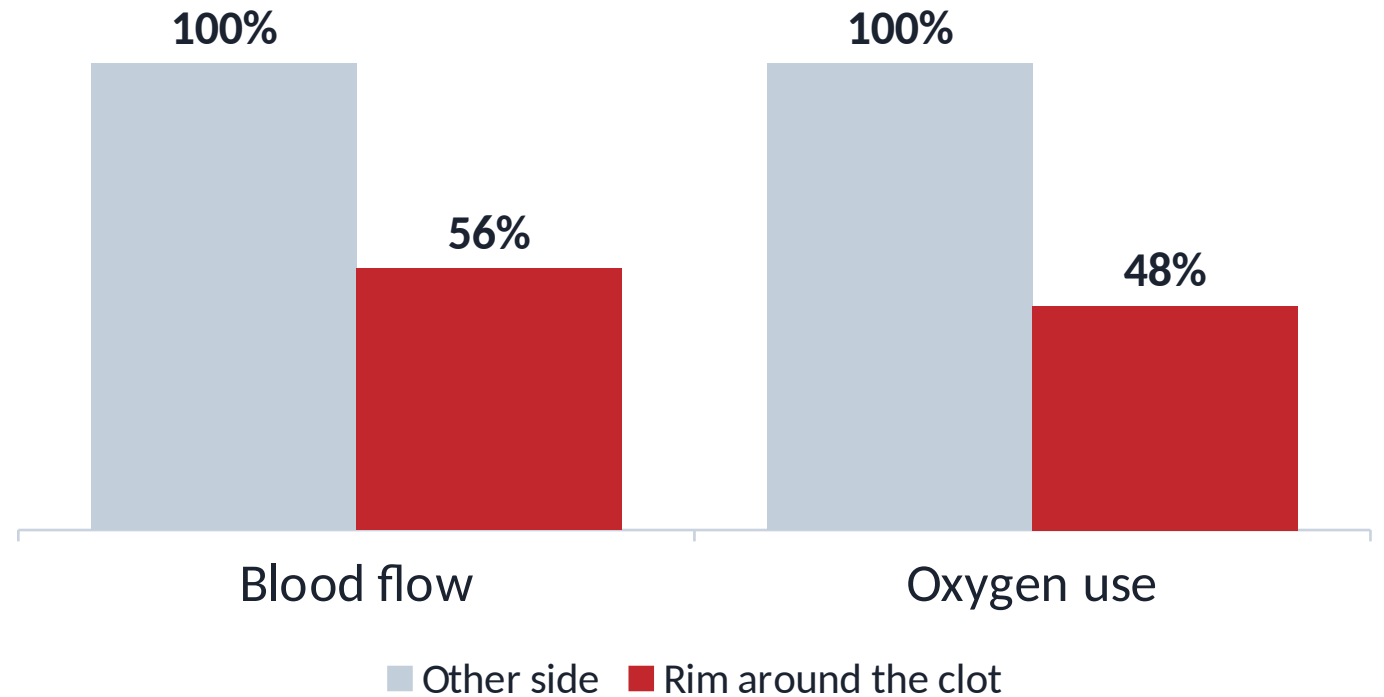
“The rim around the clot is starving tissue.”

False: the rim is resting, not starving

“The rim around the clot is starving tissue.”

Lowering BP below 150 did not reduce flow around the clot (ICH ADAPT).

Rim vs the same area on the other side (PET)



PET, n = 19, 5–22 h after onset (Zazulia et al., 2001); ICH ADAPT randomised trial, n = 75 (Butcher et al., 2013).

04

At the bedside

What it means for physiotherapy



Early helps; very early and intense may not

AVERT Trial Collaboration Group (2015) • RCT • n = 2104 • infarcts and bleeds

High dose within 24 h: fewer good outcomes (46% vs 50%)

Liu et al. (2014) • RCT • n = 243 • ICH

Rehab started within 48 h: better MBI, fewer deaths

Yen et al. (2020) • RCT • n = 60 • mild to moderate ICH

Out of bed at 24–72 h: better FIM-motor and walking



Short and frequent sessions did better (Bernhardt et al., 2016)

Certainty: low (Kan et al., 2026)

Kan et al. (2026): 17 RCTs, n = 1396. Blood pressure variability was unchanged with early mobilisation (Yen et al., 2021).

Day 4: drowsier than yesterday

MR K, 68 • A COMPOSITE CASE

Left putaminal bleed, 22 mL

Long-standing hypertension

Yesterday: sat out 20 min, GCS 15

TODAY, 9 A.M.

- **GCS 13**
- **New headache**
- **Vomited once**
- **BP 184/98**

Planned: sit-to-stand practice. What do you do?

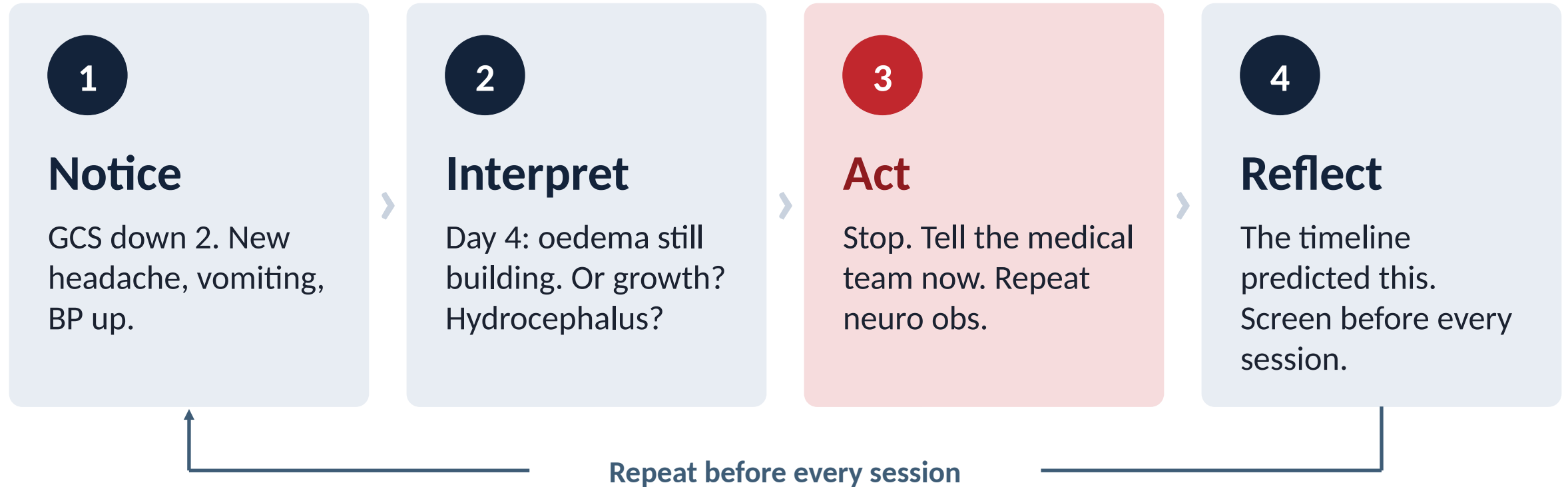
A Go ahead, but shorter

B Bed-level work only

C Hold and escalate

A composite teaching case, not a real patient.

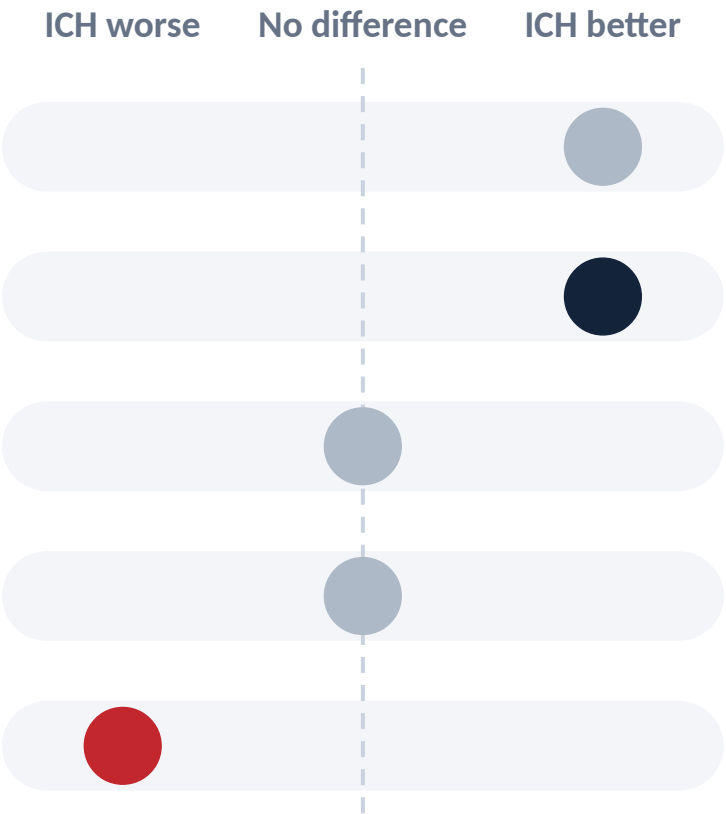
C: hold the session and escalate



Clinical reasoning cycle adapted from Levett-Jones et al. (2010).

It depends what you match

- Kelly et al. (2003)
n = 1064 • unmatched
- Paolucci et al. (2003)
n = 270 • matched on the deficit
- Salvadori et al. (2021)
n = 229 • unmatched
- Oosterveer et al. (2022)
n = 360 • unmatched
- Balk et al. (2026)
n = 194 • matched on lesion size, site



MATCH ON THE DEFICIT

ICH looks better

More of the deficit is pressure, and pressure resolves.

MATCH ON THE LESION

ICH looks worse

The blood itself is toxic.

Each dot shows the direction of the main finding, not its size. Severe early deficit is not a fixed ceiling (Yoo & Chung, 2026).

Every stage is a window to act



The vessel
Before and after

Know the cause. Check BP before, during and after activity.



The first hours
First 24–72 hours

Neuro obs first. Short, gentle sessions; nothing intense.



The days after
Days to weeks

Screen every session. Expect ups and downs. Drowsy: escalate.



The months after
Weeks to months

No early ceiling. Target what the location predicts.

(AVERT Trial Collaboration Group, 2015; Bernhardt et al., 2016; Levett-Jones et al., 2010; Puig et al., 2019; Yoo & Chung, 2026)

If you remember three things

1

The vessel decides where it bleeds.

2

The first hours decide how big it gets.

3

The blood keeps injuring the brain for days.

Adapted from Rossi and Cordonnier (2014).

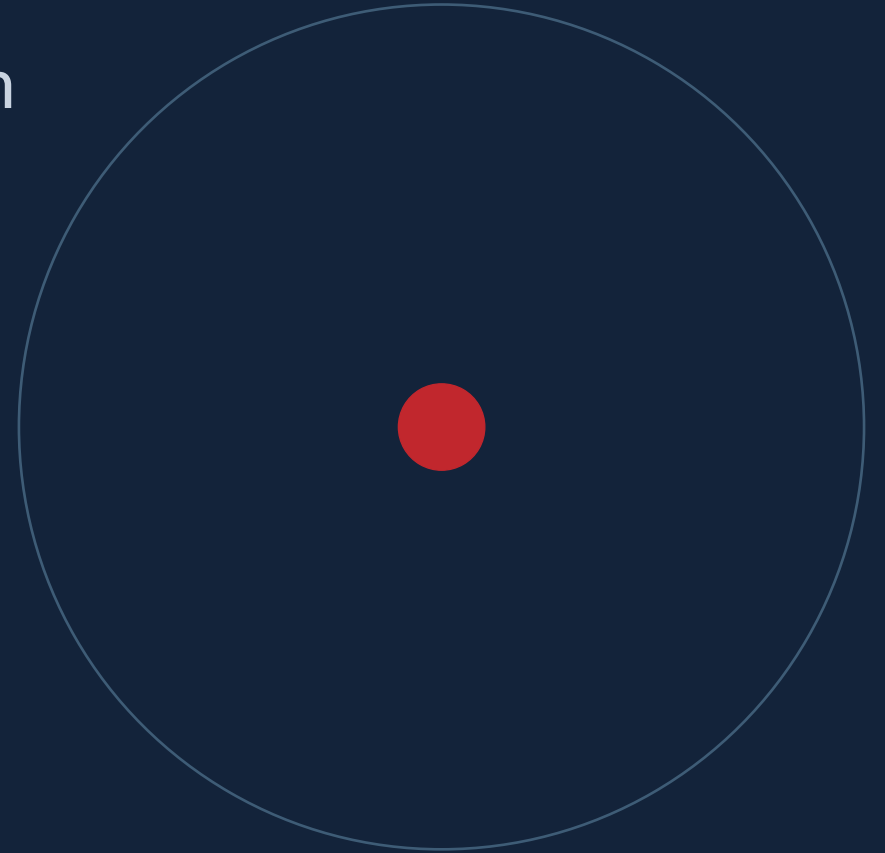
EIGHT YEARS LATER

Jill walked onto the TED stage holding a human brain.

**ICH is a process,
not a moment.**

Which claim would you challenge?

Source: Taylor (2008a, 2008b).



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